ASWATHI S CAPSTONE PROJECT

DATA CLEANING, EDA STATISTICAL ANALYSIS AND DATA STORAGE

Data Acquisition (Web Scraping):

- I used Beautiful Soup to parse the HTML table from the E-Commerce website .
- Extracted columns: Company, Revenue, Price, Country.
- Stored them in a pandas Data Frame for structured handling by using google colab with python.

Data Cleaning:

- **Numeric Cleaning:**
  - Removed symbols ($, commas, T, B, M).
  - Converted values into floats with multipliers (trillions, billions, millions).
- **Country Cleaning:**
  - Standardized names (e.g., “United States” → “USA”, “China” → “CHINA”).
  - Removed unwanted characters.
- **Missing Values:**
  - Dropped rows with NaN to ensure clean analysis.

Data Storage:

- After cleaning, the dataset is stored in a pandas Data Frame (df).
- This acts like a structured database table in memory.
- I export it to CSV/Excel

EDA

Statistical Analysis

With clean numeric data, we can run correlations

Example:

- Correlation between Rank and Revenue (Pearson coefficient).
- Classification (SVM) to distinguish countries based on revenue/price.
- Descriptive Statistical : Mean, median, Variance of revenues.

MODELS

K MEANS

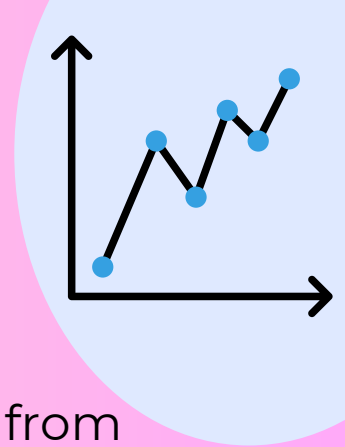
Unsupervised:
KNN classifies a new company by looking at the “k” closest companies in this dataset (neighbors) and assigning the majority class among them.
Why it fits here:
My scraped dataset is numeric (Revenue, Price), so distance-based methods like KNN are easy to find
It adapts to the local structure of the data.
It’s useful for small datasets where interpretability and simplicity matter.

SVM

- I Scraped 97 companies, but filtered down to USA/China for binary classification model
- SVM helped test whether financial metrics could distinguish companies by country.
- Hyperparameter tuning (C, gamma, kernel) ensured the model wasn’t just fitted, but optimized for accuracy.

LR

Logistic Regression:
Applied sigmoid (S-curve) to map financial features into probabilities for binary classification.



LOGISTIC REGRESSION

- **Scraping**: Company rank, name, and market capital values from the website.
- **Cleaning**: Converted market cap strings (like $691B) into numeric values.
- **Target creation**: Defined a binary label High_MarketCap → 1 if a company's market cap is above the median, 0 otherwise.
- **Feature**: Used Rank as the predictor
- **Logistic Regression**: Trained a supervised model to estimate the probability that a company is "high market cap" given its rank.
- **S-curve plot**: Showed how probability smoothly transitions from near 1 (top-ranked companies) to near 0 (lower-ranked companies).

Hyperparameter Tuning

Purpose

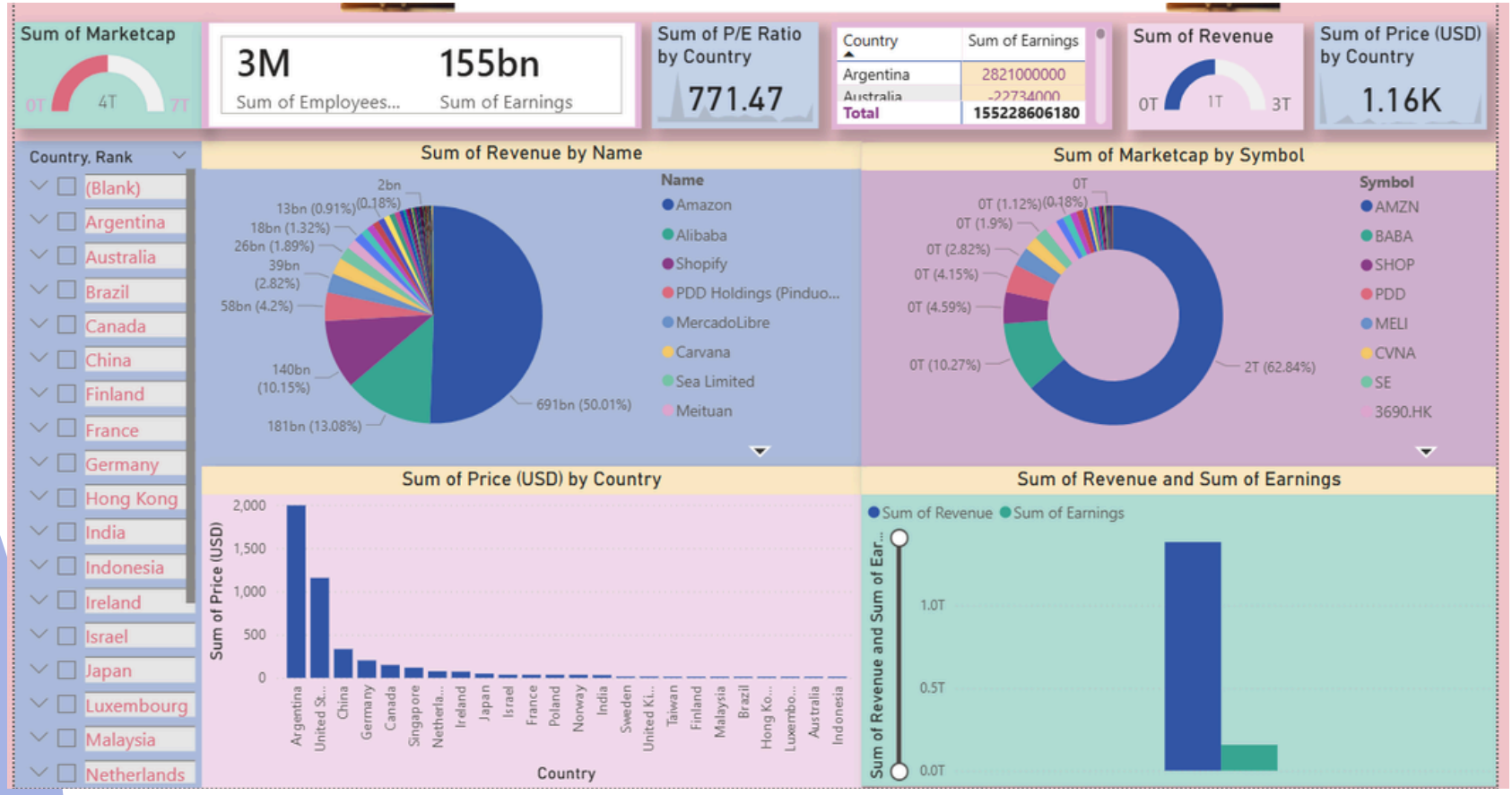
Hyperparameter tuning is about finding the best combination of these settings so the model balances accuracy, generalization, and robustness.

For this web scrap data, tuning helps avoid overfitting to extreme values and ensures the classifier works well across all companies.

In short

Hyperparameters are the “knobs” we adjust before training, and tuning them is how you optimize our model’s performance.

DASHBOARD



A Power BI dashboard is a visual report that combines multiple charts, tables, and KPIs into one interactive view. The dashboard shows metrics like Revenue, Earnings, Employees, P/E Ratio, and Price by Country, helping to quickly compare performance across regions and companies.